



Review of Literature

Progress in Proposed research /specific area

**Title Implementing Smart Contracts Using Blockchain
Technology for Secure and Transparent Legal
Documentation**

Presented By :

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Ph.D Scholar

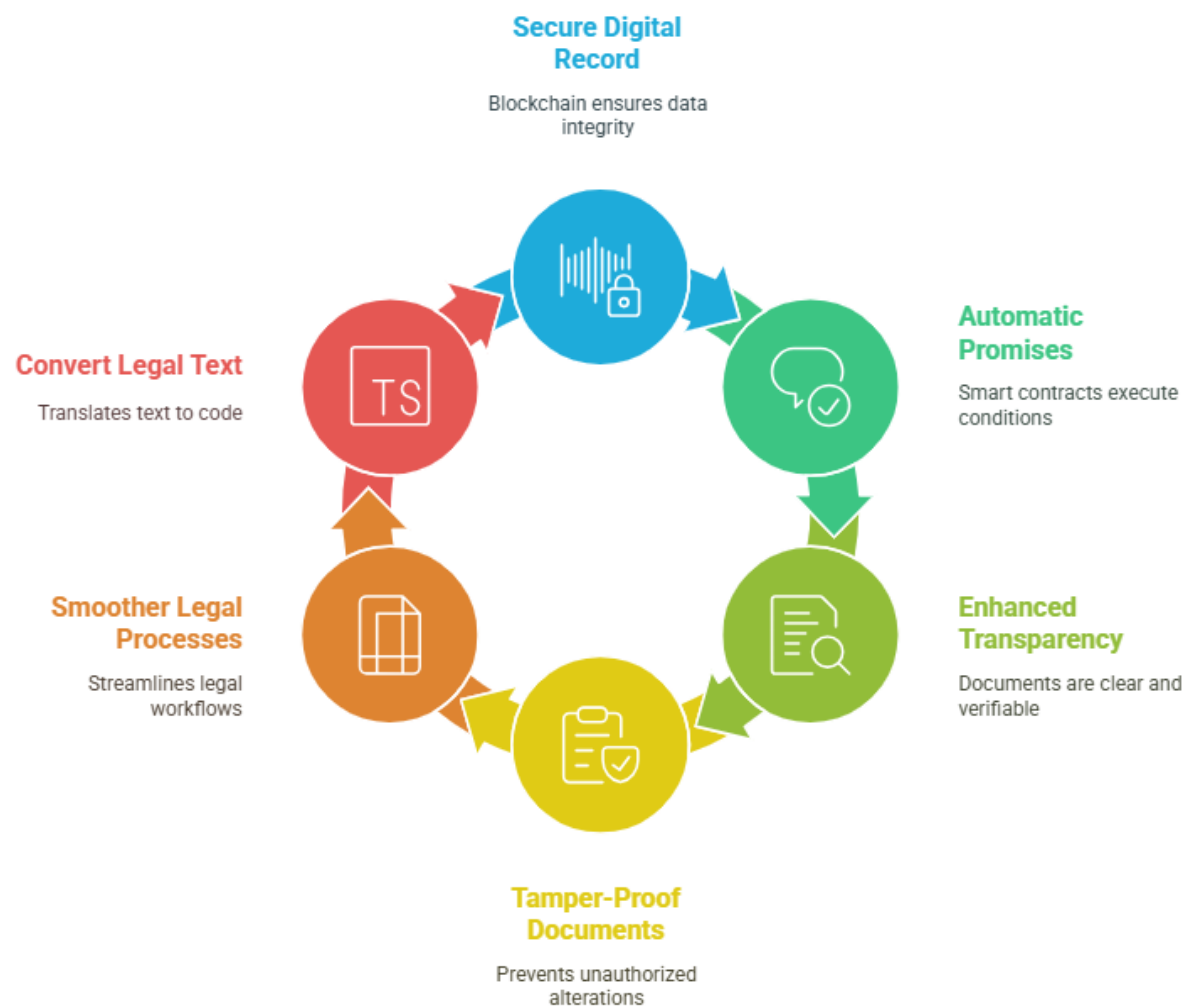
(Computer Application)

Introduction: About Topic

- The landscape of legal documentation, despite its critical importance, remains largely characterized by manual, often inefficient, and opaque processes, which are susceptible to errors and disputes.
- While the advent of blockchain and smart contracts offers a transformative promise of enhanced security, transparency, and automation, their practical integration into legal workflows is frequently hindered by a critical oversight: the human element.
- My research directly addresses this challenge by proposing a human-centered framework. To systematically achieve this, my study is guided by four distinct objectives:
- First, we aim to investigate the core principles of smart contracts and blockchain technology, specifically focusing on their nuanced application within legal documentation.
- Second, we will evaluate the security features inherent in blockchain to ensure the integrity and transparency of smart contracts, thereby enhancing their reliability for critical legal purposes.
- Third, and central to this work, is to design and propose a comprehensive framework for implementing smart contracts in legal documentation, meticulously addressing both their technical intricacies and practical usability aspects.
- Finally, we will measure the effectiveness of this framework in tangible terms: reducing costs, improving efficiency, and minimizing fraud within legal documentation processes.
- Through these objectives, this research seeks to deliver a robust, user-centric solution that bridges the gap between technological potential and practical legal application, ultimately fostering a more secure, transparent, and accessible legal future.

The Technological Promise: Blockchain & Smart Contracts

Blockchain and Smart Contract Integration Cycle



The Research Gap: Technology Isn't Enough

- Existing solutions focus on the technology, not the people.
- They often ignore the workflow of lawyers and the needs of clients.
- This leads to a critical gap: a lack of a human-centered framework.

Aim and Objectives

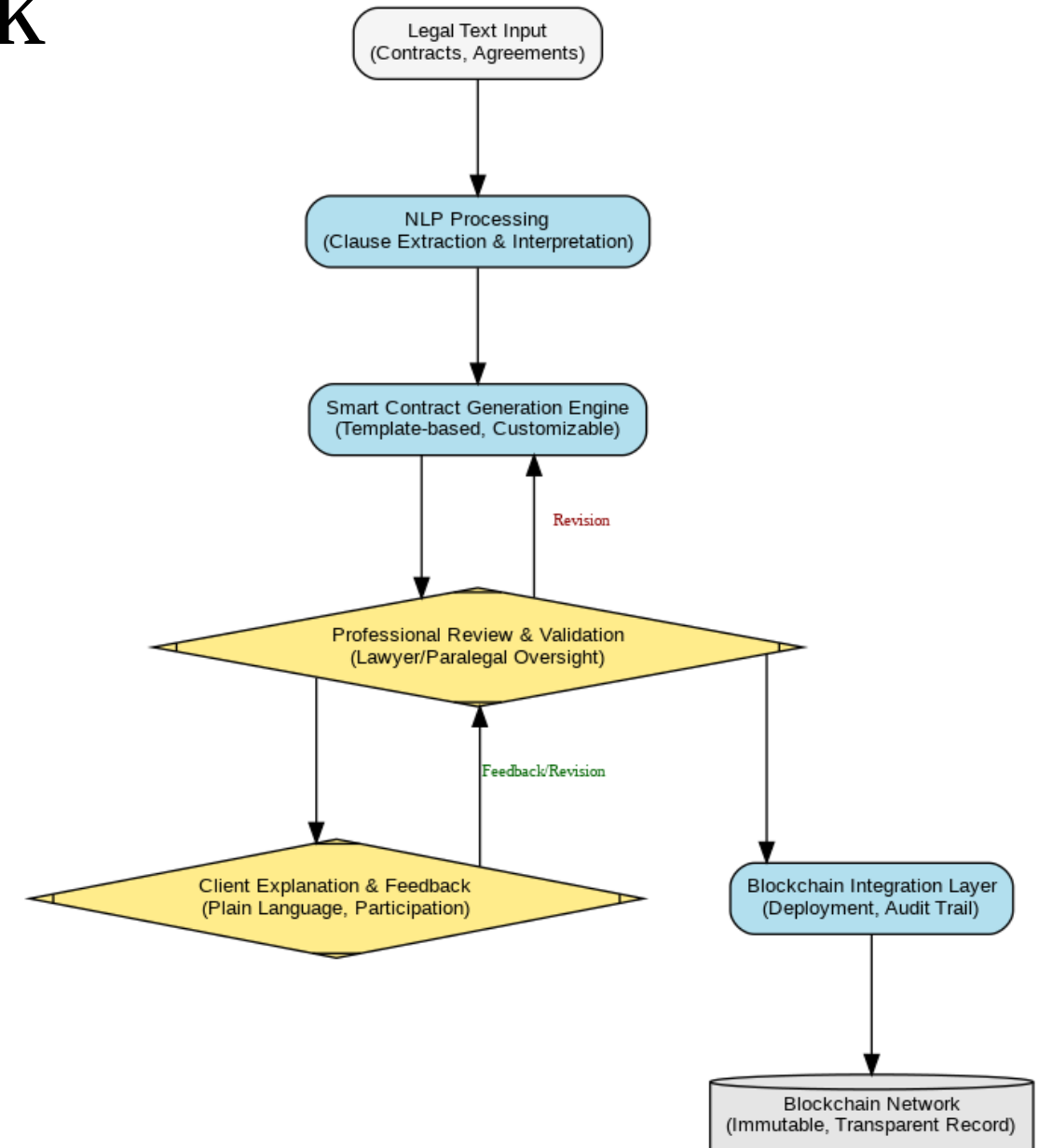
- Aim: To implement a human-centered framework for creating secure and transparent legal documentation using smart contracts and blockchain.
- Objectives:
 - Investigate principles of smart contracts & blockchain for legal use.
 - Evaluate security features for legal reliability.
 - Design and Propose a comprehensive, human-centered framework.
 - Measure its effectiveness in reducing costs, improving efficiency, and minimizing fraud.

My Approach: Design Science Research (DSR)

- This research uses a practical Design Science Research (DSR) approach.
- The focus is on building and evaluating a real-world artifact (the framework) through an iterative cycle:
- Build -> Evaluate -> Learn -> Repeat

The Proposed Framework Architecture

- Legal documents are fed to an AI that extracts clauses and generates a smart contract.
- Before deployment, legal professionals and clients review and approve a plain-language version.
- Only after this human validation is the contract securely published to the blockchain, creating an immutable, automatically-executing agreement.
- This process ensures secure, transparent, and user-friendly legal documentation.



How We Will Measure Progress

- Technical Metrics:
 - Accuracy (Clause Extraction)
 - Efficiency (Time Saved)
 - Security (Vulnerability Analysis)
- Human-Centered Metrics:
 - User Satisfaction (SUS Score)
 - Learning Curve (Time to Proficiency)
 - Client Understanding & Accessibility

Expected Contributions

- **For Academia**: A novel, empirically-validated, human-centered framework for smart contract generation.
- **For Legal Professionals**: A tool to improve efficiency, reduce errors, and enhance client communication.
- **For Society**: Increased transparency, trust, and access to justice in legal processes.

Review of Literature

Problem: Traditional legal documentation processes are slow, opaque, and susceptible to errors and fraud. Existing technological solutions often fail because they ignore the human element—the needs of legal professionals and the comprehension challenges for clients.

Research Gap: There is a lack of a comprehensive, human-centered framework that integrates advanced NLP and blockchain technology to make legal processes secure, transparent, and user-friendly.

Aim & Objectives: The research aims to design, build, and evaluate such a human-centered framework. The objectives are to investigate the core technologies, evaluate security features, design the framework, and measure its real-world effectiveness.

Methodology: The project will use Design Science Research (DSR) combined with agile principles. It involves a multi-phase approach:

understanding the problem, designing the solution collaboratively, building and testing prototypes, and finally, evaluating the framework in real-world settings.

Proposed Framework: A four-part architecture is proposed:

1. A Human-AI Collaboration Interface for legal professionals.
2. An Intelligent Document Processing Module using NLP.
3. A template-based Smart Contract Generation Engine with human oversight.
4. A Blockchain Integration Layer for secure and transparent deployment.

Evaluation: Success will be measured using both technical metrics (accuracy, efficiency, security) and crucial human-centered metrics (user satisfaction, learning curve, client understanding, and access to justice).

Objective	Mapped Paper	Justification for Mapping	Key Finding/Summary	Limitation/Critical Review
Objective 1: Investigate core principles of smart contracts and blockchain for legal documentation	Szabo, J., et al. - Computer Life Legal Implications and Challenges of Blockchain Technology and Smart Contracts	Directly addresses the application and challenges of the technology within the legal domain.	Discusses legal and regulatory hurdles for the adoption of blockchain and smart contracts, including enforceability, jurisdiction, and the translation of legal	Frames the problem from a legal theory perspective. The research bridges this theory with practical, human-centered software implementation.
	Nakamoto, S. - Bitcoin: A Peer-to-Peer Electronic Cash System	Establishes the fundamental principles of decentralized, peer-to-peer ledger technology.	The foundational whitepaper that introduced the concept of a decentralized, peer-to-peer electronic cash system, which formed the basis for blockchain	This is the historical and technical origin of the core technology. It is cited to establish the foundation and has no 'limitation' in this context, as it created the field.
	Sultan, K., et al. - CONCEPTUALIZING BLOCKCHAINS: CHARACTERISTICS & APPLICATIONS	Defines the core characteristics of blockchain (immutability, transparency, etc.) that are foundational to the research.	Provides a framework for understanding blockchain's core characteristics (e.g., decentralization, immutability, transparency) and its broad	The application focus is broad, which justifies the need for a deep-dive into a specific domain like law, which this paper does not provide.
	Bassan, F., & Rabitti, M. - From smart legal contracts to contracts on blockchain: An empirical investigation	Provides an empirical investigation into the practical application of this technology to legal documents.	Provides an empirical look at the practical challenges and nuances of translating legal agreements into blockchain-based contracts, highlighting the gap between theory	Crucial for supporting the 'Research Gap' argument. It provides evidence that real-world, human-centric factors are often overlooked in purely technical implementations.
	Salah, K., et al. - Blockchain for AI: Review and open research challenges	Provides a high-level academic context for your work, which integrates AI (NLP) with Blockchain.	Reviews the intersection of Blockchain and AI, outlining synergies and research challenges.	Validates your research area as a significant and active field. Its broad scope justifies your specific focus on a human-centered legal framework.
	Bartoletti, M., et al. - Smart contract languages: A comparative analysis	Investigates the core principles and technical specifics of smart contract programming languages.	Analyzes and compares different programming languages used for smart contracts (e.g., Solidity, Vyper).	Informs the technical implementation details of the framework, justifying the choice of a specific language and its implications for security and usability.
	Lei, Z. - Application Of Smart Contracts Under the Framework of Contract Law Introduction	Analyzes how the technology fits within existing legal doctrine, a key part of the investigation.	Analyzes how smart contracts fit within the existing legal framework of contract law.	Complements the other legal papers by providing a direct analysis of the intersection between the technology and established legal doctrine.

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Objective 2: Evaluate security features for reliability in legal contexts	Nelaturu, K., et al. - Correct-by-Design Interacting Smart Contracts...with VeriSolid	Addresses reliability by proposing a method to ensure contracts are built correctly from the start (verification).	Presents a formal verification approach to ensure that smart contracts are 'correct-by-design' before deployment.	This is a key reference for the 'Verification' and 'Professional Review' stages of the architecture. It shows the importance of formal methods.
	Mnasri, M., et al. - A systematic literature review on security testing of Ethereum smart contracts	Provides a comprehensive overview of methods to evaluate the security of smart contracts.	Reviews the existing literature on security testing for smart contracts, categorizing tools and techniques.	Essential for 'Evaluation' and 'Security Metrics'. This paper provides the basis for the types of security analyses you plan to conduct.
	Liang, R., et al. - Vulseye: Detect Smart Contract Vulnerabilities via Stateful Directed Graybox Fuzzing	Details a specific, state-of-the-art technique for evaluating security, showing awareness of current methods.	Introduces an advanced technique (fuzzing) for detecting vulnerabilities in smart contracts.	A specific example of a state-of-the-art security testing tool you can reference in your evaluation methodology.
	Siddiqui, S., et al. - A smart-contract-based adaptive security governance architecture for smart city service interoperations	Provides a parallel for the legal domain, which is also a complex system with security needs.	Proposes a security architecture for smart cities, an environment with many interacting, complex components.	The principles of security governance for a complex smart city can be adapted for the governance of your legal framework.
	Guarnizo, J., & Szalachowski, P. - PDFS: Practical Data Feed Service for Smart Contracts	Addresses the 'oracle problem' which is a key security and reliability challenge for contracts needing external data.	Addresses the 'oracle problem' of how smart contracts can securely and reliably get information from the outside world.	Relevant for the future work section. While the framework focuses on on-chain logic, interacting with real-world data would require a secure oracle.
	Wei, G., et al. - Consolidating Smart Contracts with Behavioral Contracts	Supports the goal of ensuring the generated contract accurately reflects the 'intent' of the legal agreement.	Proposes using 'behavioral contracts' to specify and verify the expected behavior of smart contracts over time.	This is a strong theoretical paper that supports the goal of ensuring reliability by matching implementation to intent.
	Marmsoler, D., & Thornton, B. - Deductive verification of solidity smart contracts with SS Calc	Presents a formal method for evaluating the correctness and security of contracts.	Presents a tool and method for formally proving that a Solidity smart contract meets its specification.	Another key paper for the 'Verification' part of the methodology. It provides a specific example of a formal verification technique.
	Peelam, M. S., et al. - Decentralized Trust: NFT and Blockchain-Enabled Evidence System using Fog Computing	Explores how blockchain ensures integrity and trust for legal evidence, a direct parallel to legal agreements.	Proposes a system for managing evidence in a decentralized and trustworthy manner.	Highly relevant. This framework creates trustworthy agreements; this paper discusses how to manage trustworthy evidence, a natural extension.
	Bedi, P., et al. - DBESN: A novel model for detecting and identifying malicious code in a smart contract	Focuses on proactive security evaluation by detecting malicious code before deployment.	Presents a model specifically for detecting malicious code within smart contracts before they are deployed.	Another strong citation for the security evaluation methodology, focusing on proactive threat detection.

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Objective 3: Design and propose a comprehensive framework	Zou, W., et al. - Smart Contract Development: Challenges and Opportunities	Justifies the need for a new framework by outlining the challenges that the proposed design aims to solve.	Identifies key challenges in the smart contract development lifecycle, including security, performance, and correctness.	It provides a solution to the usability and domain-specific challenges that this paper identifies as open problems.
	Ahmed, S. U., et al. - Smart Contract Generation through NLP and Blockchain for Legal Documents	Directly informs the design of the NLP module for interpreting legal text.	Proposes a method for generating smart contracts from legal text using NLP.	It differentiates by adding a human-in-the-loop validation and client feedback mechanism, which this paper does not focus on.
	Fang, P., et al. - iSyn: Semi-automated Smart Contract Synthesis from Legal Financial Agreements	Supports the Human-AI Collaboration design, showing that a semi-automated approach is practical.	Presents a semi-automated system for synthesizing smart contracts, requiring human intervention for complex tasks.	Provides evidence that a fully automated approach is not yet feasible, making the human-in-the-loop approach more practical and robust.
	Skotnica, M., et al. - Towards Model-Driven Smart Contract Systems	Informs the design of the template-based Generation Engine.	Advocates for a model-driven engineering approach where contracts are generated from higher-level conceptual models.	This is a form of model-driven engineering, which justifies your approach of not having lawyers code from scratch but rather work with pre-approved templates.
	Marchesi, L., et al. - A blockchain architecture for industrial applications	Provides a reference architecture that informs the design of the framework's Blockchain Integration Layer.	Details a reference architecture for applying blockchain in industrial settings.	Shows that your proposed architecture is based on established principles for real-world, industrial-grade applications.
	Pan, Y., et al. - Automated Generation of Security-Centric Descriptions for Smart Contract Bytecode	Supports the design of the client explanation and professional review features.	Focuses on automatically creating human-readable descriptions of what a smart contract does, specifically for security analysis.	Highlights the importance of translating complex code into understandable descriptions, a core tenet of your human-centered design.
	Alzahrani, A., et al. - Design and implementation of a decentralized trustless data standardization framework for blockchain interoperability	Informs the future potential and scalability of the framework design.	Focuses on creating standards so that different blockchain systems can work together.	Relevant for the 'future work' or 'scalability' discussion. A standardized legal smart contract framework could interoperate with other systems.
	Kutbi, M. - Named Entity Recognition Utilized to Enhance Text Classification While Preserving Privacy	Provides a specific, citable technique for the NLP module's design.	Details a specific NLP technique (NER) for text analysis, which is a core part of your proposed NLP module.	This paper provides a concrete example of the technology that will be used in the 'Clause Extraction & Interpretation' phase of your architecture.
	El-Samad, W., et al. - AI - Driven Data Aggregation Level Smart Contracts for Blockchain Healthcare Insurance Claims Adjudication	Provides a parallel design from the insurance domain, validating the overall approach.	Demonstrates using AI and smart contracts to automate insurance claims, another document-heavy and rule-based domain.	Excellent parallel to the legal field. Insurance claims, like legal contracts, are based on specific conditions and require high trust. This paper validates the approach in a similar domain.

Objective	Mapped Paper	Justification for Mapping	Key Finding/Summary	Limitation/Critical Review
Objective 4: Measure the effectiveness of the framework	Emami, A., et al. - A blockchain-driven business model for supplier selection	Provides a model for how to measure effectiveness by linking technical features to outcomes.	Proposes a model using blockchain for transparent supplier selection, linking technical features to measurable business outcomes like efficiency and risk reduction.	This paper is a great example for the 'Metrics' section. It shows how to connect a technical implementation to tangible, measurable value.
	Al-Fuqaha, A., et al. - A pilot study on developing smart contracts in the construction industry	Serves as a model for how to conduct a pilot study to empirically evaluate the framework's effectiveness.	Examines the practical application of smart contracts in the Korean construction industry, identifying domain-specific needs.	This is an excellent example of a domain-specific study. It can use be to argue that just as construction has unique needs, the legal domain requires its own specialized framework and evaluation.
	Huang, W., et al. - Smart blockchain-powered natural resource asset management and ecological governance countermeasures	Demonstrates how the effectiveness of a blockchain system can be measured in a different complex domain.	Shows the application of blockchain for managing assets and governance in a completely different field (environmental science).	Provides a strong parallel application. If blockchain can be used to govern natural resources, it can be adapted to govern legal agreements, and its effectiveness can be measured similarly.
	Roushan, A., et al. - A multi-objective supply chain model for disaster relief optimization	Shows the application and implied effectiveness in a high-stakes environment.	Applies smart contracts to a high-stakes, complex environment (disaster relief), showing their utility beyond simple financial transactions.	Demonstrates the robustness of smart contracts for complex, multi-party situations, which is analogous to the high stakes of legal agreements and provides a model for evaluating effectiveness in such contexts.
	Nasser, A., et al. - Blockchain technology in maritime single window and port community systems	The methodology of a systematic review itself is a way to measure the current state of the art.	A systematic review of blockchain applications in the maritime industry, another complex logistical domain.	This paper's methodology (a systematic literature review) serves as a model for your own literature review chapter, which is the baseline against which the effectiveness of a new framework is judged.

Conclusion

- This research addresses a critical need at the intersection of law and technology.
- By prioritizing a human-centered design approach, the proposed framework moves beyond a purely technical implementation of smart contracts.
- It aims to create a practical, usable, and trustworthy tool that empowers legal professionals and demystifies complex legal agreements for clients.
- The expected contribution is a significant step towards a more efficient, transparent, and accessible legal system, demonstrating that the true potential of blockchain technology is realized only when it is designed to serve human needs.
- This work seeks to provide a robust, empirically-validated model for the future of secure digital legal documentation.

SNO.	Author'S Name & Affiliation	Title of paper/ Article/chapter	NAME OF JOURNAL/ Book with Publisher	YEAR Vol, Issue	Finding/summary	Critical Review
1	Zou, W., et al.	Smart Contract Development: Challenges and Opportunities	IEEE Transactions on Software Engineering	2021	Identifies key challenges in the smart contract development lifecycle, including security, performance, and correctness.	Justifies the need for a structured, human-centered framework by outlining the very problems your research addresses. Your work provides a solution to the usability and domain- specific challenges it mentions.
2	Szabo, J., et al.	Computer Life Legal Implications and Challenges of Blockchain Technology and Smart Contracts	Emerging Industrial Applications: Orchestration of Machine Learning, the IoT, and Blockchain	2024	Discusses legal hurdles for blockchain adoption, including enforceability, jurisdiction, and translating legal principles into code.	Provides the essential legal context for your thesis. Your research bridges the gap between the legal theory it discusses and the practical software engineering required.
3	Nakamoto, S.	Bitcoin: A Peer-to-Peer Electronic Cash System	www.bitcoin.org	2008	The foundational whitepaper that introduced the decentralized, peer- to-peer electronic cash system, forming the basis for blockchain.	The historical and technical origin of the core technology. Foundational citation for the introduction.

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4	Sultan, K., et al.	CONCEPTUALIZING BLOCKCHAINS: CHARACTERISTICS & APPLICATIONS	arXiv	2018	Provides a framework for understanding block chain's core characteristics (decentralization, immutability, transparency) and its broad applications.	It defines the core technology. The broad application focus justifies your deep-dive into the specific legal domain.
5	Bassan, F., & Rabitti, M.	From smart legal contracts to contracts on blockchain: An empirical investigation	Computer Law and Security Review	2024	Provides an empirical look at the practical challenges of translating legal agreements into blockchain contracts.	Crucial for supporting your 'Research Gap' argument. It provides evidence that real-world, human-centric factors are often overlooked.
6	Ahmed, S. U., et al.	Smart Contract Generation through NLP and Blockchain for Legal Documents	Procedia Computer Science	2024	Proposes a method for generating smart contracts from legal text using NLP.	Directly relevant to your NLP module. Your work differentiates by adding a human-in-the-loop validation and client feedback mechanism.

SNO.	Author'S Name & Affiliation	Title of paper/ Article/chapter	NAME OF JOURNAL/ Book with Publisher	YEAR Vol, Issue	Finding/summary	Critical Review
7	Fang, P., et al.	iSyn: Semi-automated Smart Contract Synthesis from Legal Financial Agreements	ISSTA 2023	2023	Presents a semi-automated system for synthesizing smart contracts, requiring human intervention for complex tasks.	Supports your Human-AI collaboration model. It provides evidence that a fully automated approach is not yet feasible, making your approach more practical.
8	Nelaturu, K., et al.	Correct-by-Design Interacting Smart Contracts...with VeriSolid	IEEE Trans. Dependable Secure Comput.	2023	Presents a formal verification approach to ensure that smart contracts are 'correct-by-design' before deployment.	Key reference for the 'Verification' and 'Professional Review' stages of your architecture. It shows the importance of formal methods.
9	Skotnica, M., et al.	Towards Model-Driven Smart Contract Systems...	2018 48th Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshops	2020	Advocates for a model-driven engineering approach where contracts are generated from higher-level conceptual models.	Supports the 'template-based' aspect of your Generation Engine, justifying not having lawyers code from scratch.

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10	Marchesi, L., et al.	A blockchain architecture for industrial applications	Blockchain: Research and Applications	2022	Details a reference architecture for applying blockchain in industrial settings.	Provides a solid architectural foundation for your 'Blockchain Integration Layer', showing it's based on established principles.
11	Mnasri, M., et al.	A systematic literature review on security testing of Ethereum smart contracts	Blockchain: Research and Applications	2025	Reviews the existing literature on security testing for smart contracts, categorizing tools and techniques.	Essential for your 'Evaluation' chapter and 'Security Metrics'. This paper provides the basis for the types of security analyses you plan to conduct.
12	Liang, R., et al.	Vulseye: Detect Smart Contract Vulnerabilities via Stateful Directed Graybox Fuzzing	IEEE TIFS	2025	Introduces an advanced technique (fuzzing) for detecting vulnerabilities in smart contracts.	A specific example of a state-of-the-art security testing tool you can reference in your evaluation methodology.

SNO.	Author'S Name & Affiliation	Title of paper/ Article/chapter	NAME OF JOURNAL/ Book with Publisher	YEAR Vol, Issue	Finding/summary	Critical Review
13	Emami, A., et al.	A blockchain-driven business model for supplier selection...	Digital Business	2025	Proposes a model using blockchain for transparent supplier selection, linking technical features to measurable business outcomes.	Excellent example for Metrics section. It shows how to connect a technical implementation to tangible, measurable value.
14	Salah, K., et al.	Blockchain for AI: Review and open research challenges	IEEE Access	2019	Reviews the intersection of Blockchain and AI, outlining synergies and research challenges.	Provides a high-level academic context for your work, which integrates AI (NLP) with Blockchain. It validates your research area as a significant and active field.
15	Bartoletti, M., et al.	Smart contract languages: A comparative analysis	Future Generation Computer Systems	2025	Analyzes and compares different programming languages used for smart contracts (e.g., Solidity, Vyper).	Informs the technical implementation details of your framework, justifying the choice of a specific language and its implications for security and usability.

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16	Siddiqui, S., et al.	A smart-contract-based adaptive security governance architecture for smart city service interoperations	Sustainable Cities and Society	2024	Proposes a security architecture for smart cities, an environment with many interacting, complex components.	Provides a parallel for the legal domain, which is also a complex system. The principles of security governance here can be adapted for your framework.
17	Guarnizo, J., & Szalachowski, P.	PDFS: Practical Data Feed Service for Smart Contracts	arXiv	2019	Addresses the 'oracle problem' of how smart contracts can securely and reliably get information from the outside world.	Relevant for the future work section of your thesis. While your framework focuses on on-chain logic, interacting with real-world data (e.g., a court ruling) would require a secure oracle.
18	Pan, Y., et al.	Automated Generation of Security-Centric Descriptions for Smart Contract Bytecode	ISSTA 2023	2023	Focuses on automatically creating human-readable descriptions of what a smart contract does, specifically for security analysis.	Supports the 'client explanation' and 'professional review' aspects of your framework. It shows the importance of translating complex code into understandable descriptions.

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19	Wei, G., et al.	Consolidating Smart Contracts with Behavioral Contracts	Proc. ACM Program. Lang.	2024	Proposes using 'behavioral contracts' to specify and verify the expected behavior of smart contracts over time.	This is a strong theoretical paper that supports your goal of ensuring the generated contract accurately reflects the 'intent' of the legal agreement.
20	Marmsoler, D., & Thornton, B.	Deductive verification of solidity smart contracts with SSCalc	Sci. Comput. Program.	2025	Presents a tool and method for formally proving that a Solidity smart contract meets its specification.	Another key paper for the 'Verification' part of your methodology. It provides a specific example of a formal verification technique you could theoretically integrate.
21	Al-Fuqaha, A., et al.	A pilot study on developing smart contracts in the construction industry...	KSCE Journal of Civil Engineering	2025	Examines the practical application of smart contracts in the Korean construction industry, identifying domain-specific needs.	Excellent example of a domain-specific study. You can use it to argue that just as construction has unique needs, the legal domain requires its own specialized, human-centered framework.

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22	Alzahrani, A., et al.	Design and implementation of a decentralized trustless data standardization framework for blockchain interoperability...	Alexandria Engineering Journal	2025	Focuses on creating standards so that different blockchain systems can work together.	Relevant for the 'future work' or 'scalability' discussion. A standardized legal smart contract framework could interoperate with other systems (e.g., government registries).
23	Huang, W., et al.	Smart blockchain-powered natural resource asset management and ecological governance countermeasures	Heliyon	2025	Shows the application of blockchain for managing assets and governance in a completely different field (environmental science).	Provides a strong parallel application. If blockchain can be used to govern natural resources, it can certainly be adapted to govern legal agreements.
24	Kutbi, M.	Named Entity Recognition Utilized to Enhance Text Classification While Preserving Privacy	IEEE Access	2023	Details a specific NLP technique (NER) for text analysis, which is a core part of your proposed NLP module.	Provides a specific, citable technique for the 'Clause Extraction & Interpretation' phase of your architecture.

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25	El-Samad, W., et al.	AI -Driven Data Aggregation Level Smart Contracts for Blockchain Healthcare Insurance Claims Adjudication	Procedia Computer Science	2024	Demonstrates using AI and smart contracts to automate insurance claims, another document-heavy and rule-based domain.	Excellent parallel to the legal field. Insurance claims, like legal contracts, are based on specific conditions and require high trust. This paper validates the approach in a similar domain.
26	Lei, Z.	Application Of Smart Contracts Under the Framework of Contract Law Introduction	Journal of Education Humanities and Social Sciences 35:226-233	2024	Analyzes how smart contracts fit within the existing legal framework of contract law.	Complements the other legal papers by providing a direct analysis of the intersection between the technology and established legal doctrine.
27	Peelam, M. S., et al.	Decentralized Trust: NFT and Blockchain-Enabled Evidence System using Fog Computing	Blockchain: Research and Applications	2025	Proposes a system for managing evidence in a decentralized and trustworthy manner.	Highly relevant. Your framework creates trustworthy agreements; this paper discusses how to manage trustworthy evidence, a natural extension for legal applications.

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28	Bedi, P., et al.	DBESN: A novel model for detecting and identifying malicious code in a smart contract	Blockchain: Research and Applications	2025	Presents a model specifically for detecting malicious code within smart contracts before they are deployed.	Another strong citation for your security evaluation methodology, focusing on proactive threat detection.
29	Roushan, A., et al.	A multi-objective supply chain model for disaster relief optimization using neutrosophic programming and blockchain-based smart contracts	Supply Chain Analytics	2025	Applies smart contracts to a high-stakes, complex environment (disaster relief), showing their utility beyond simple financial transactions.	Demonstrates the robustness of smart contracts for complex, multi-party situations, which is analogous to complex legal agreements.
30	Nasser, A., et al.	Blockchain technology in maritime single window and port community systems: A bibliometric analysis and systematic literature review	Telematics and Informatics Reports Volume 18, June 2025, 100206	2025	A systematic review of blockchain applications in the maritime industry, another complex logistical domain.	This paper's methodology (a systematic literature review) serves as a model for your own literature review chapter, validating your approach.

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THANK YOU

Deepak Bhatt
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